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Clinical Overview of Pulmonary and Pleural Neoplasms 2026

Jason Golbin DO, MBA, MS

November 2026

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Case 1

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- HPI
  - 61-year-old female transferred with “massive” hemoptysis
  - PMH: HTN
  - Social history: 60 pack year smoking history

Physical exam

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- BP: 146/104, Pulse: 101, RR: 27/min, O2 sat: 92%
- No palpable lymphadenopathy
- Lungs: decreased breath sounds Left base
- Heart : S1S2 tachycardic
- Abdomen: soft, NT, + BS

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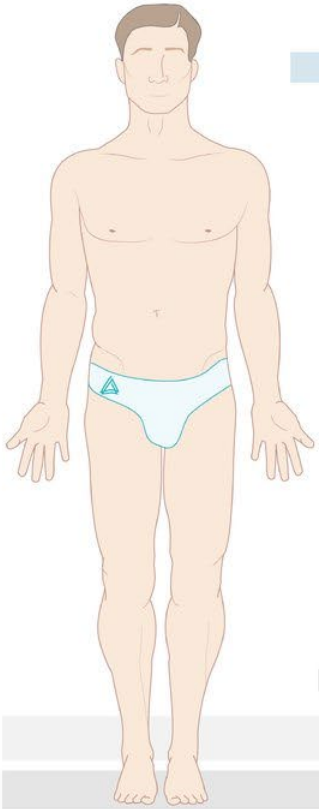
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What's our differential?

Epidemiology – Lung Cancer

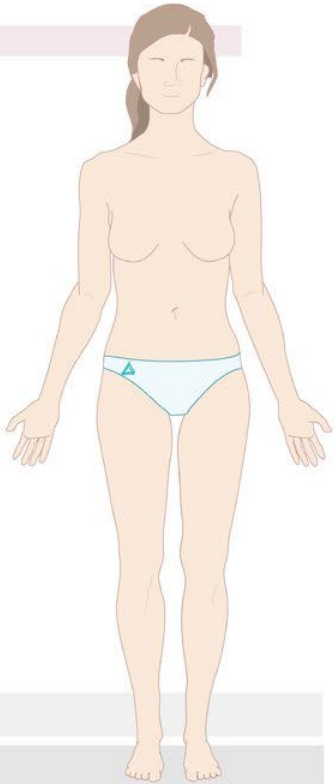
- Most common cancer worldwide
- Worldwide, over 234,000 new cases diagnosed yearly
- Leading cause of cancer-related death among men and women
  - 1.6 million deaths annually
- In US, over 154,000 lung cancer-associated deaths annually

Source: Uptodate



New cases per year

Men		Total		Women
Prostate	19.0 %	161,360	252,710	30.0 % Mammary gland
Lung & bronchus	14.0 %	116,990	105,510	12.0 % Lung & bronchus
Colon & rectum	9.0 %	71,420	64,010	8.0 % Colon & rectum
Urinary bladder	7.0 %	60,490	61,380	7.0 % Uterine corpus
Melanoma of the skin	6.0 %	52,170	42,470	5.0 % Thyroid
Kidney & renal pelvis	5.0 %	40,610	34,940	4.0 % Melanoma of the skin
Non-Hodgkin lymphoma	5.0 %	40,080	32,160	3.5 % Non-Hodgkin lymphoma
Leukemia	4.0 %	36,290	25,840	3.0 % Leukemia
Oral cavity & pharynx	4.0 %	35,720	25,700	3.0 % Pancreas
Liver& intrahepatic bile duct	3.0 %	29,200	23,380	3.0 % Kidney &. renal pelvis



Source: Amboss

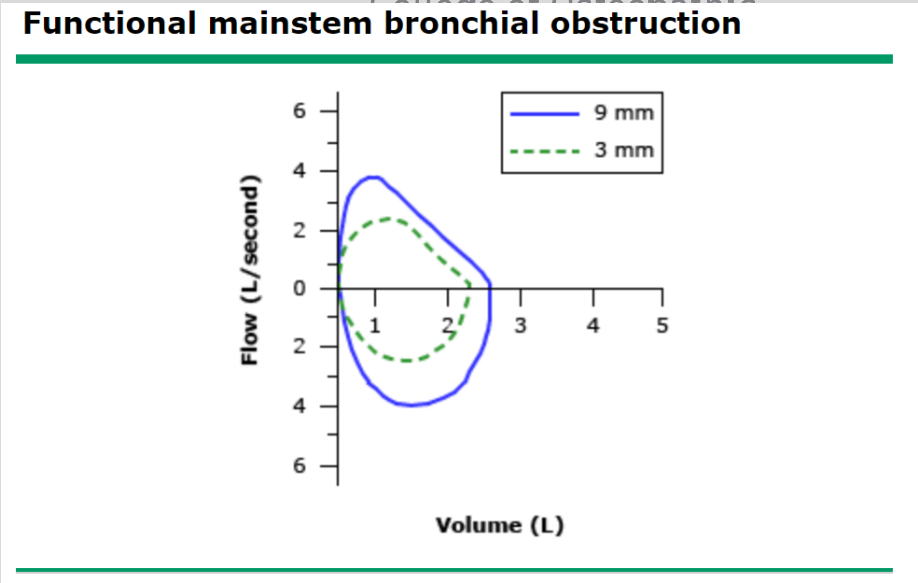
Clinical Manifestations of Lung Cancer

- **Signs and symptoms** — The majority of patients who present with clinical signs or symptoms due to lung cancer have advanced disease
- Most commonly:
  - Cough – 50 to 75%
  - Hemoptysis – 25 to 50%
  - Dyspnea – 25%
  - Chest pain – 20%
    - Pleural/mediastinal/chest wall extension
    - PE

Source: Uptodate

Clinical Manifestations of Lung Cancer

- Bronchorrhea (excessive sputum): ****seen w/ Broncho-alveolar carcinoma****
- Persistent infection
  - Failure to resolve
  - Post obstructive
  - Recurring common colds in smokers ≥ 40 years should raise the suspicion of lung cancer
- Wheeze/stridor
  - Seen in central tumors → obstructs major airways
  - Obstruction may cause abnormal PFTs/flow volume loop



Source: Uptodate

Risk Factors

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- Cigarette Smoking
  - Accounts for 90% of all lung cancers
  - 1 ppd x 40 yrs = 20x increased chance of lung Ca
  - Smoking Cessation is CRITICAL

Source: Uptodate

Risk Factors

- Radiation therapy (for other malignancies)
  - Hodgkin lymphoma, breast CA
- Environmental toxin exposure
  - secondhand smoke, asbestos, radon, arsenic, nickel, beryllium
- Pulmonary fibrosis (interstitial lung disease)
  - 7x increased risk
- HIV

Source: Uptodate

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What's the bottom line?

- Lung cancer should always be suspected in a current or former smoker with new onset of cough or hemoptysis!

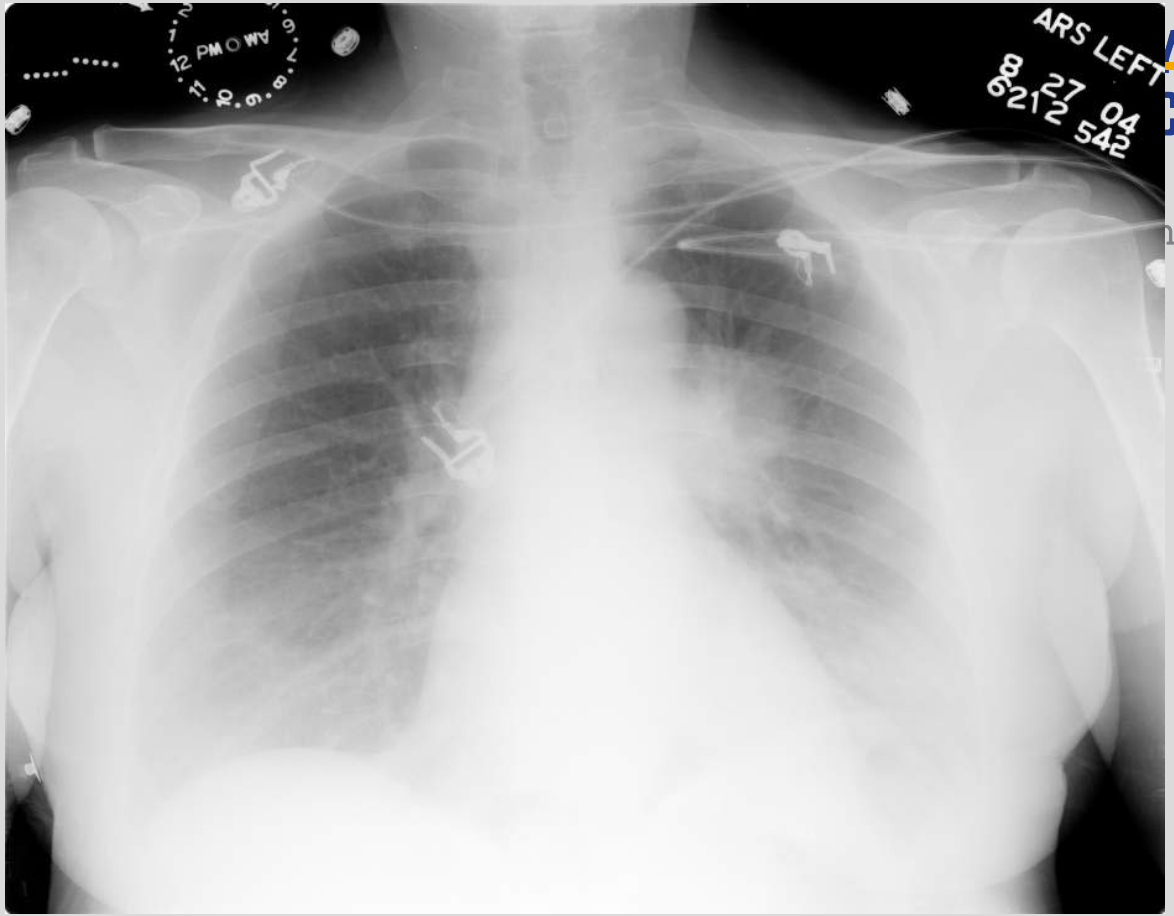
- What do you want to order????

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Diagnostics – Lung Cancer

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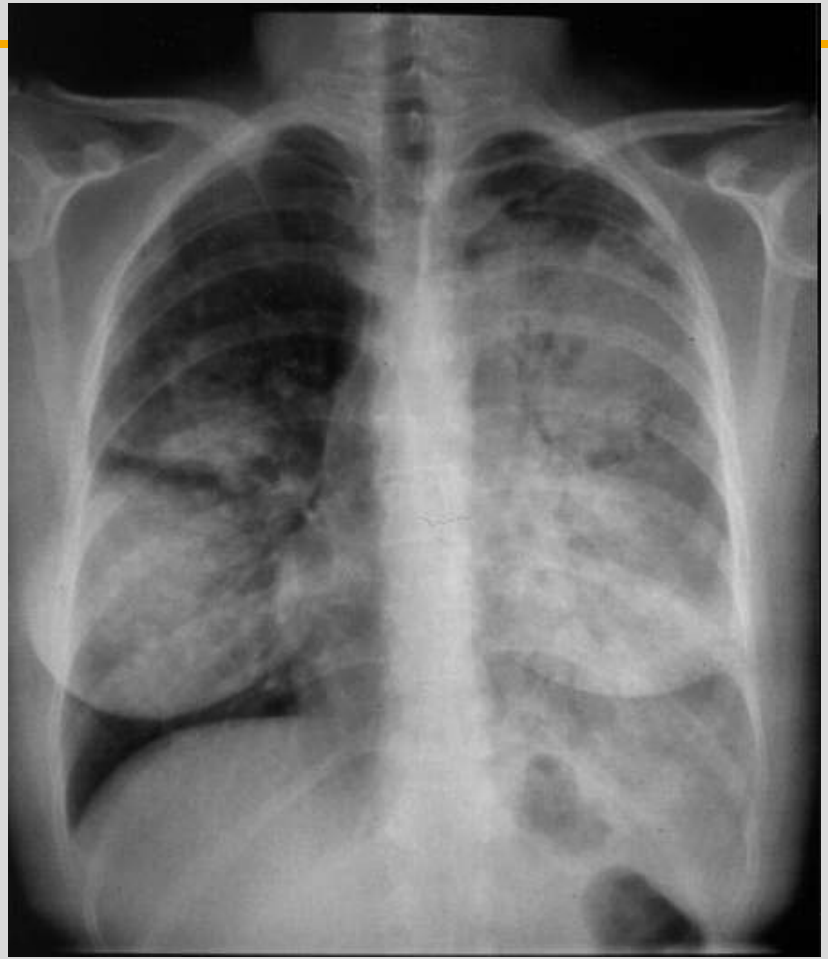
- **Chest X-ray**
- CT scan
- Bronchoscopy
  - Bronchial washings
  - Bronchial brushings
  - Bronchial biopsies
- Sputum cytology
- Percutaneous fine needle aspiration
- Thoracentesis
- Pleural biopsy
- VATS



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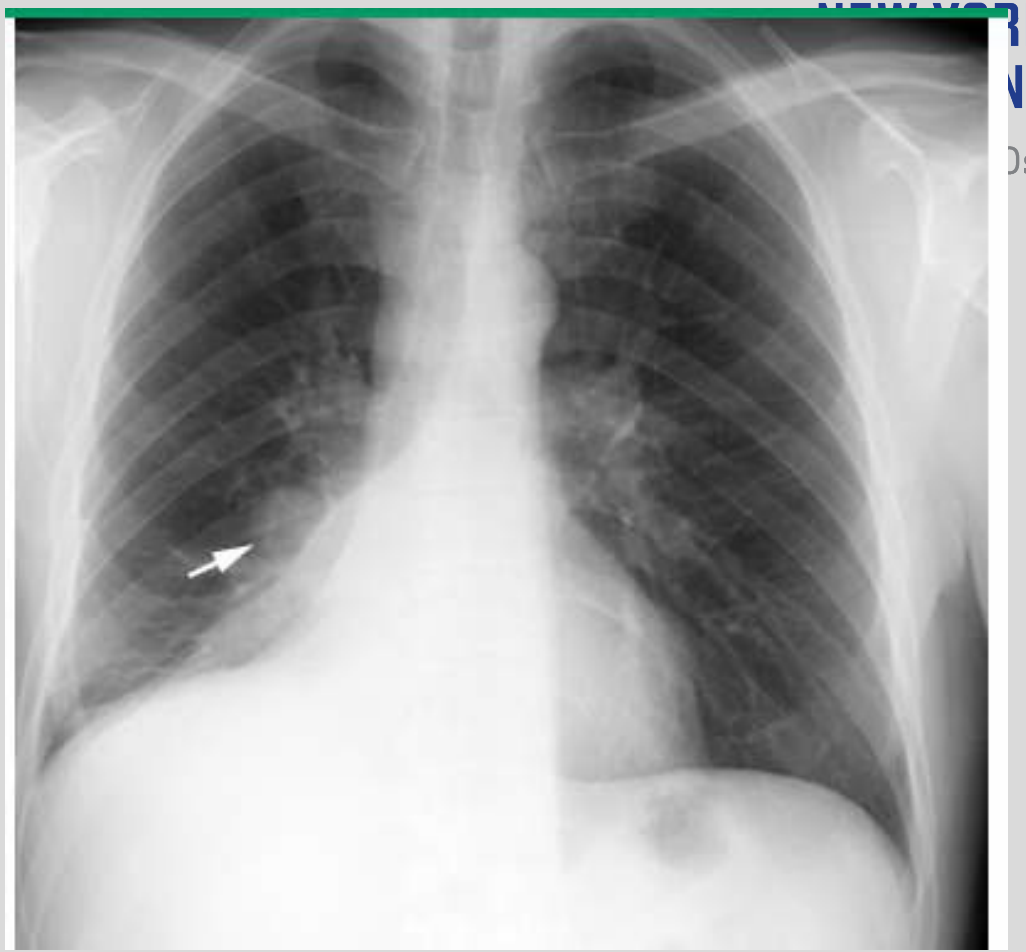


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Solitary pulmonary nodule



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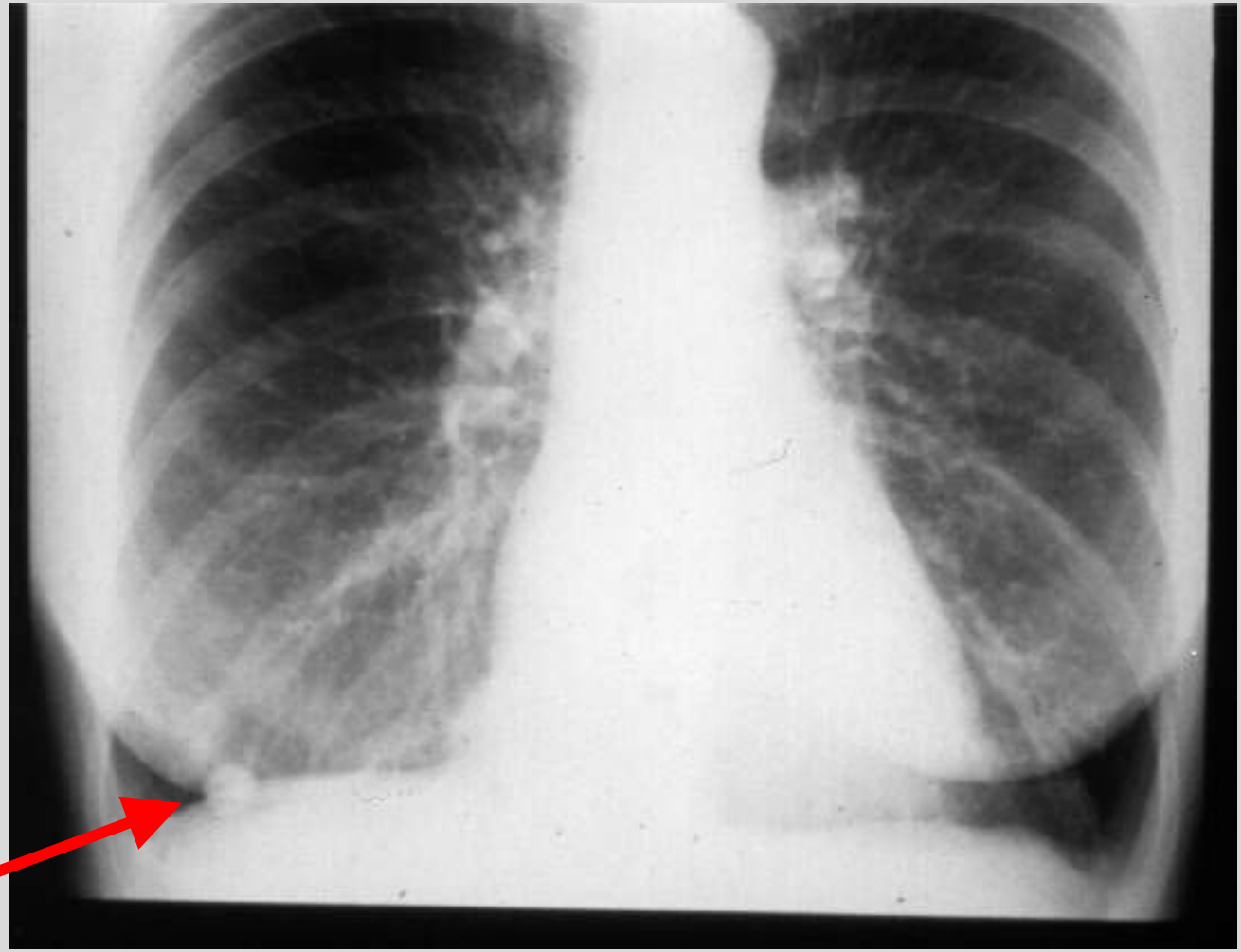
Differential of a “Solitary pulmonary nodule”

- Benign
  - Granulomatous disease
    - Mycobacteria (usually TB)
    - Fungi (histoplasmosis, coccidioidomycosis)
  - Hamartoma
    - Disorganized lung tissue and cartilage (w/ scattered calcifications)
    - “coin lesion”
- Malignant
  - Primary vs secondary

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“Coin lesion”

- Seen with hamartoma
- Benign lesion of disorganized lung tissue
- Well circumscribed
- Usually peripheral
- < 3-4 cm



Initial CXR

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- Asymptomatic patients → Lesions may be detected incidentally
- Symptomatic patients → order a chest radiograph
- ADDITIONALLY: obtain any prior chest imaging studies to determine the age and growth pattern of identified abnormalities

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What imaging do you order next?

Diagnostics – Lung Cancer

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- Chest X-ray
- **CT scan**
- Bronchoscopy
  - Bronchial washings
  - Bronchial brushings
  - Bronchial biopsies
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Computed tomography (CT) scan

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- **No previous CXR or NEW nodules/CHANGES to old nodules on previous CXR → ORDER A CT!!**
- Chest CT +/- contrast
- CT upper abdomen including liver and adrenal glands (usually by extension of the chest CT)

Computed tomography (CT) scan

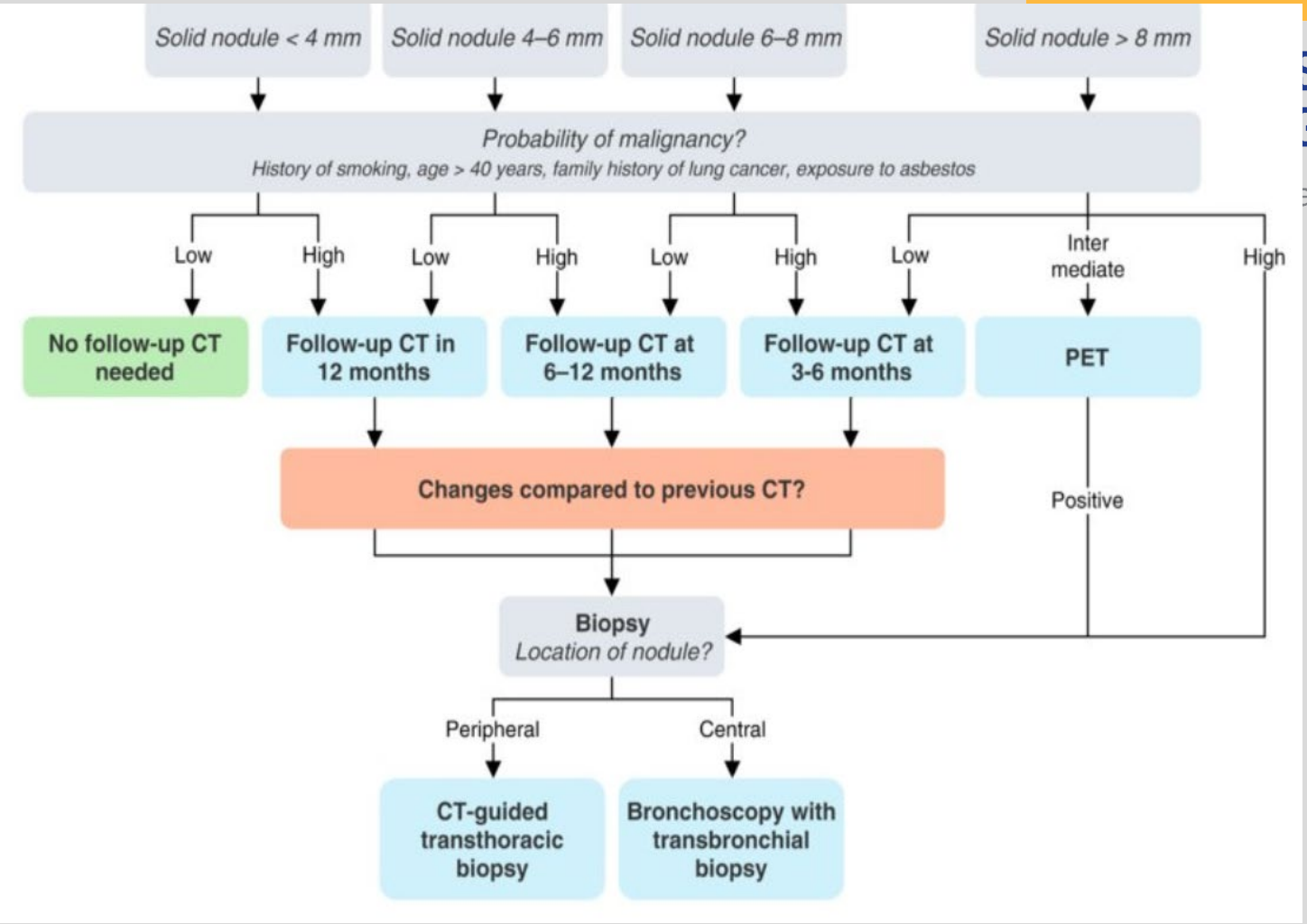
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Assess nodule:

- size (> or < 8mm)
- location (central vs peripheral)
- signs of malignancy (irregular margins, spicules, absent or irregular calcifications)

2 questions determine the next step:

1. How big is the nodule?
2. What is the probability that the nodule is malignant?
    - E.g. hx of smoking, age > 40, family hx of lung Ca, asbestos exposure, signs of malignancy, etc.


Solid nodule < 4 mm
Solid nodule 4–6 mm
Solid nodule 6–8 mm
Solid nodule > 8 mm
Probability of malignancy?
History of smoking, age > 40 years, family history of lung cancer, exposure to asbestos
Low
High
Low
High
Low
High
Low
Inter mediate
High
No follow-up CT needed
Follow-up CT in 12 months
Follow-up CT at 6–12 months
Follow-up CT at 3-6 months
PET
Changes compared to previous CT?
Positive
Biopsy
Location of nodule?
Peripheral
Central
CT-guided transthoracic biopsy
Bronchoscopy with transbronchial biopsy

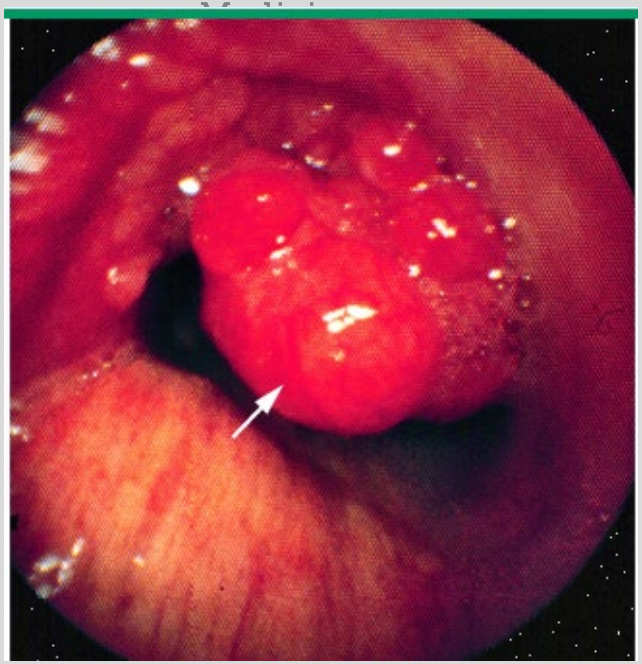
Diagnostics – Lung Cancer

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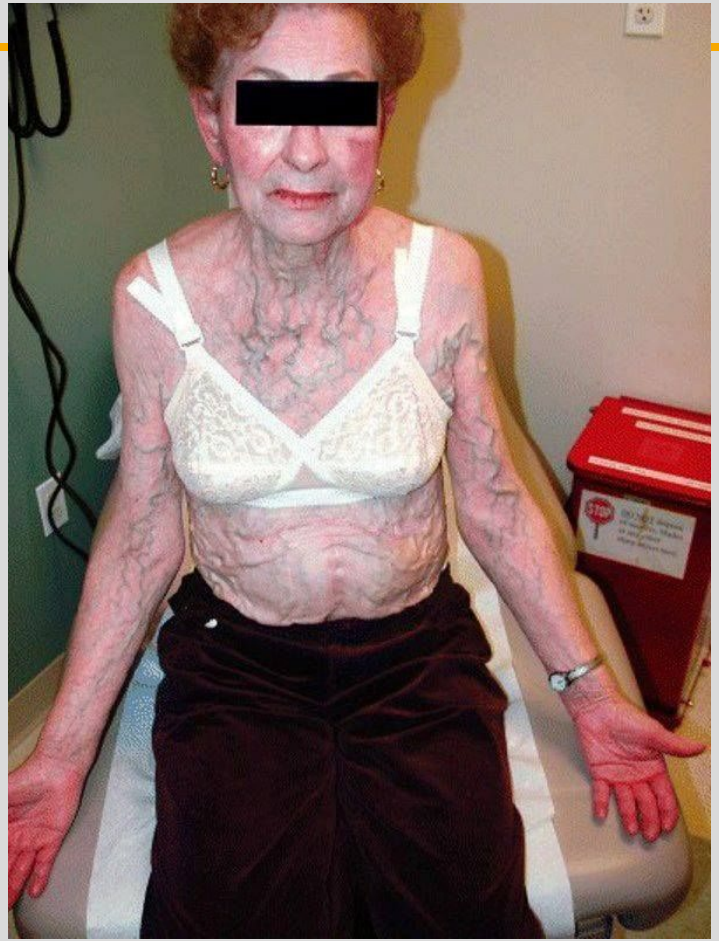
Bronchoscopy and biopsy

- Bronchoscopy with transbronchial biopsy → central nodules
  - washings
  - brushings
  - mucosal biopsy
- CT-guided transthoracic biopsy → peripheral nodules
- Major complications
  - Pneumonia
  - Pneumothorax
  - Bleeding



Case 2

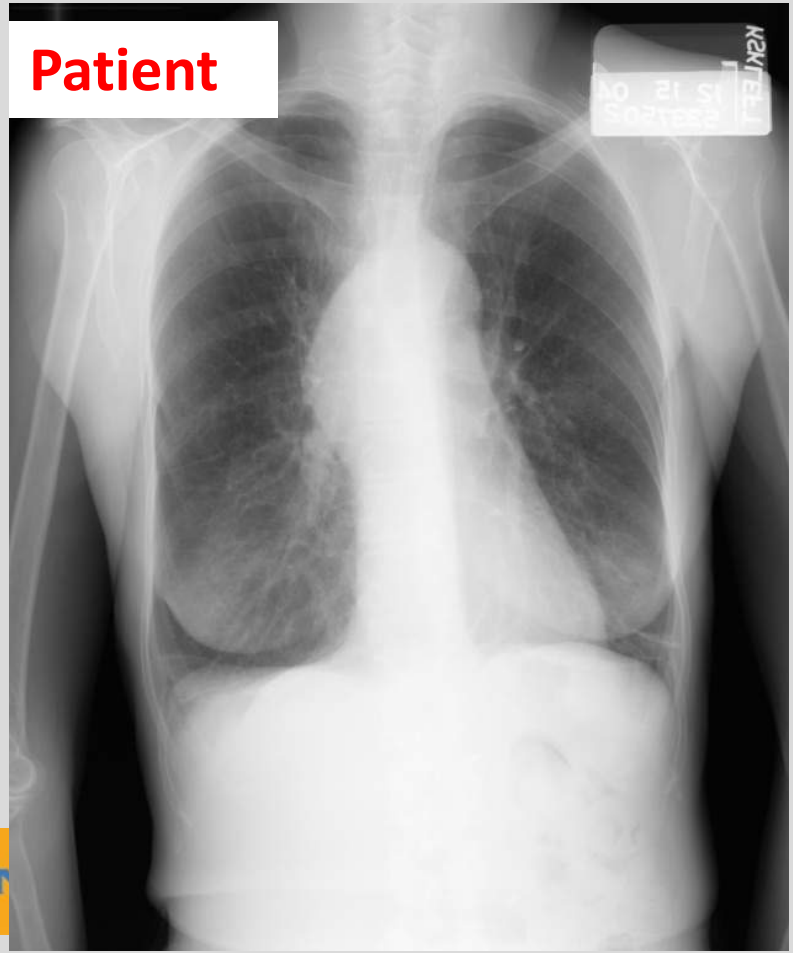
- 68 y/o Female with 5-month history of progressive dyspnea
- Also c/o fatigue, increased swelling of hands and face over the past 3 weeks
- Pt gets lightheaded when she bends over but no syncope.
- Has noted prominent veins on chest wall recently
- Labs WNL

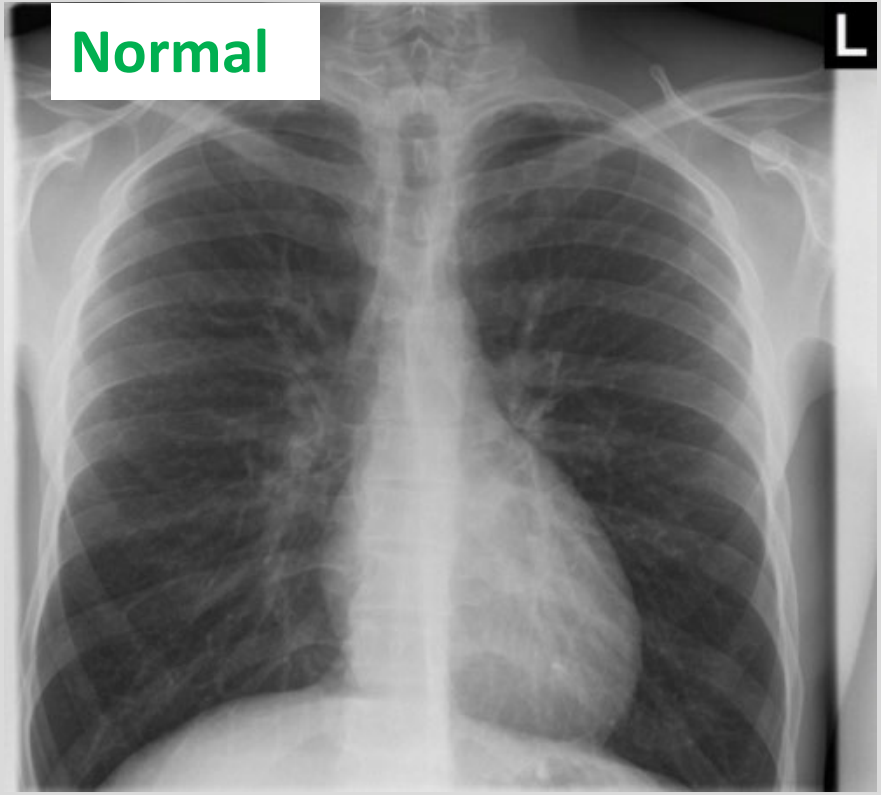


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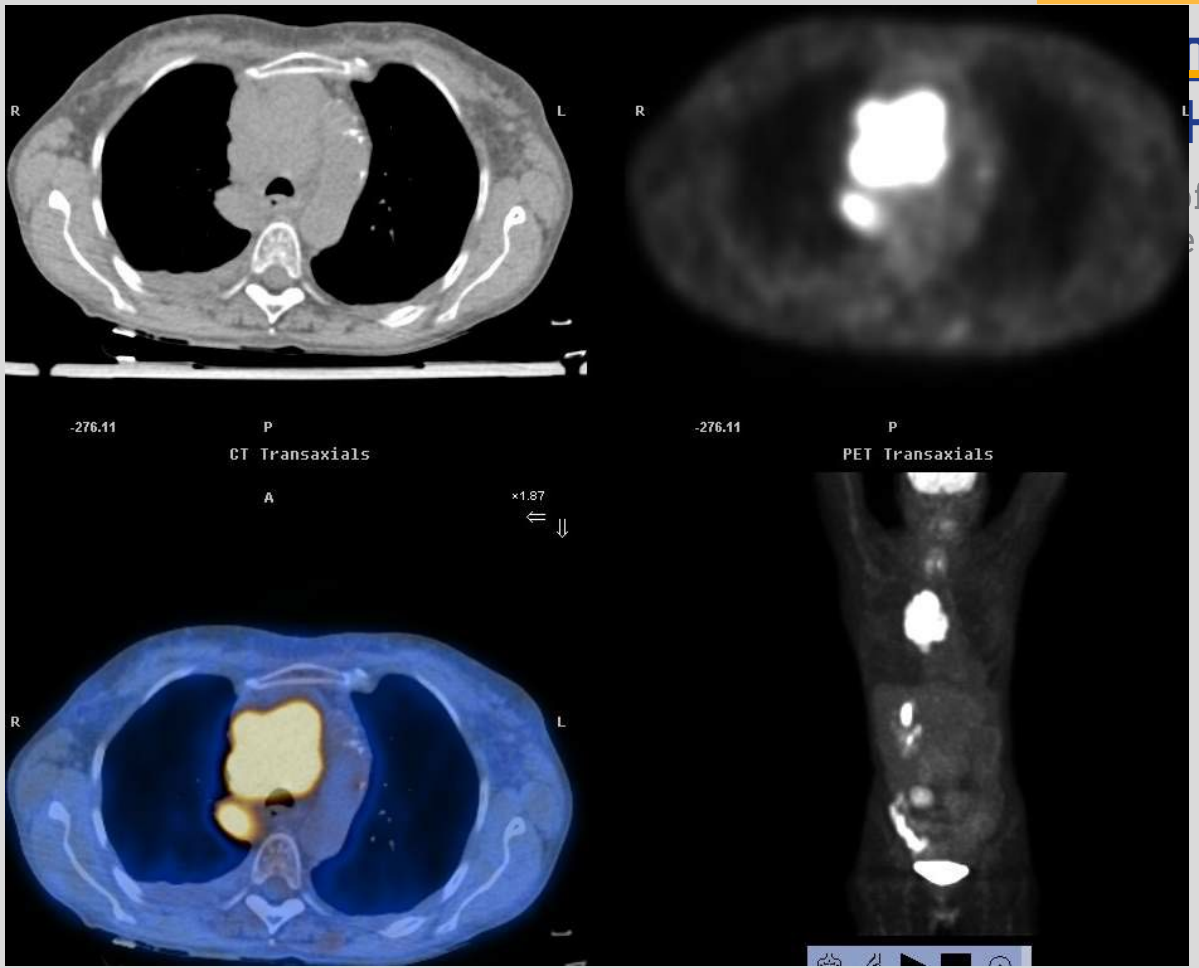

Patient


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Whole-body PET Scan

- Identifies metabolic activity of the 1° tumor, mediastinal involvement, and distant metastases.
- Controversial: not been shown to improve survival

Superior Vena Cava (SVC) Syndrome

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- Etiology
  - Compression of vena cava by tumor or mediastinal lymph nodes
  - Thrombus caused by tumor or nodal invasion
- Presentation
  - Sensation of fullness in head and dyspnea
  - Dilated neck veins
  - Prominent venous pattern on chest
  - Facial edema
  - Plethoric appearance

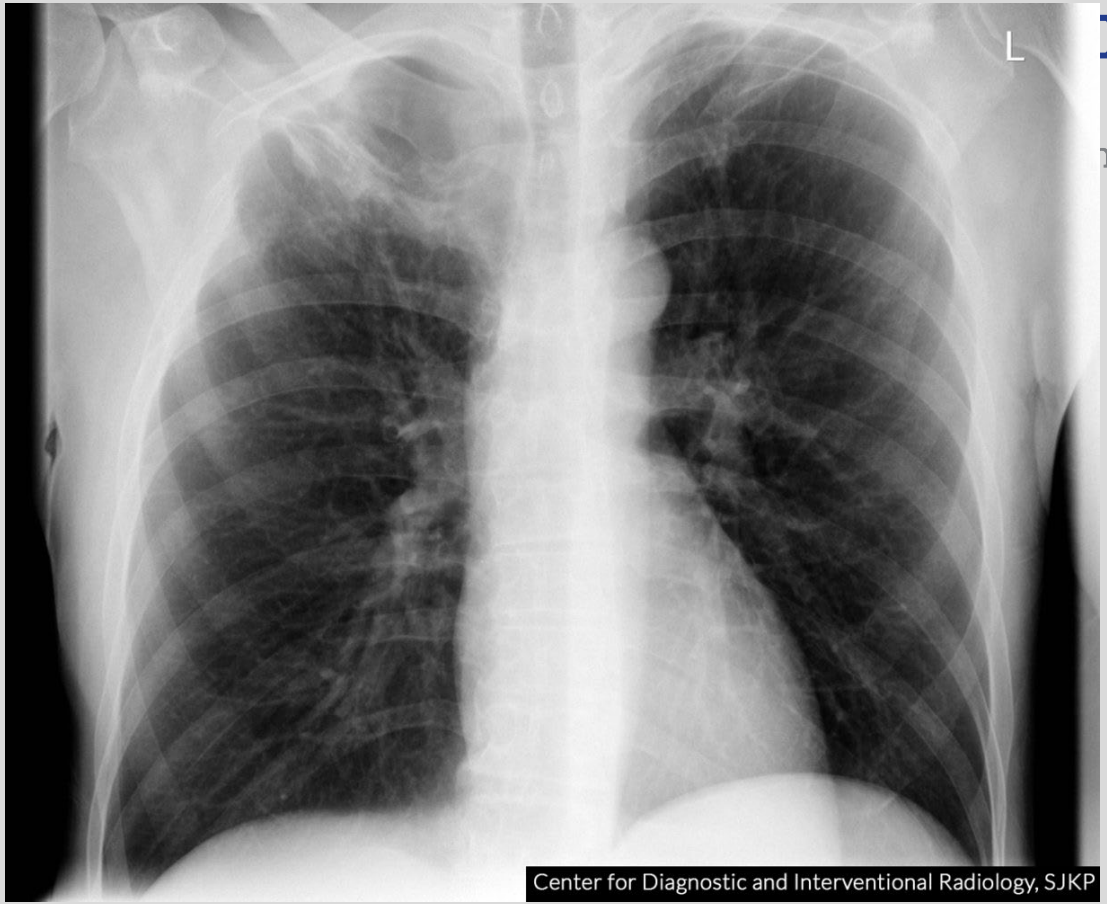
Case 3

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- 70 y/o male
- 3 months of progressive weight loss
- 2 months of U/L: shoulder pain, blurry vision, hand weakness
- 40 pk/yr smoking history



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Pancoast tumor

- Tumor involving the upper lobe in the “superior sulcus”
- Primarily non-small cell carcinoma
- Shoulder pain (*most common initial symptom*)
  - Due to invasion of brachial plexus & invasion into parietal pleura, ribs, vertebrae
- Horner’s syndrome (from paravertebral sympathetic chain invasion)
  - *Ptosis*
  - *Miosis*
  - *Anhydrosis (look for ipsilateral facial flushing)*
- Neuro-complications of UE
  - Extension into C8-T1 nerve roots → pain/weakness/atrophy of hand
    - Similar to Klumpke palsy

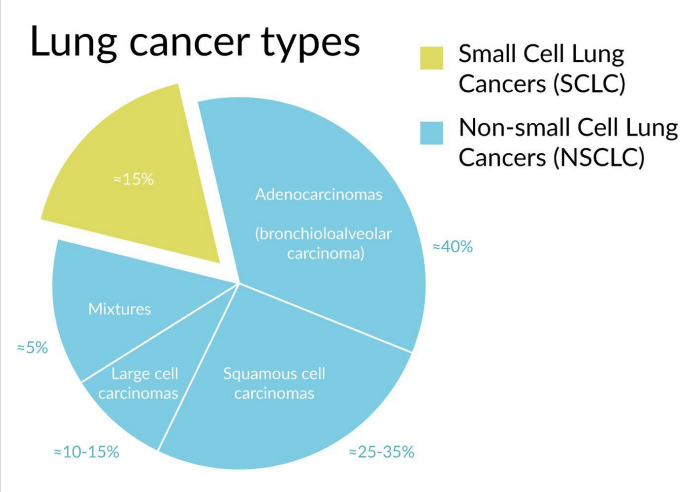
Paraneoplastic syndromes

- Hypercalcemia – PTH-related protein
  - Squamous cell
- SIADH – hyponatremia
  - Small cell
  - 10% of SCLC pts have SIADH; 75% of all cancer associated SIADH is from SCLC
- Neurologic syndromes – Lambert-Eaton most common
  - Small cell
- Cushing syndrome – ectopic ACTH
  - Small cell
- Hypertrophic Osteo-arthropathy – clubbing and periosteal proliferation of tubular bones
  - AdenoCa
- Dermatomyositis/ Polymyositis – muscle weakness

Small cell vs Non-Small Cell Lung Cancer

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- Small cell (15%)
  - Smokers only
  - Fast growing → early metastasis
  - **CENTRAL**
  - Not amenable to surgical resection → treated with chemo → poor prognosis
- Non-Small cell (85%)
  - Smokers AND non-smokers
  - Can be resected in early stages
  - Better prognosis



Source: UpToDate, Amboss

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Lung Cancer Metastasis

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- Adrenals
  - Usually found incidentally on abdominal imaging for another reason
- Brain
  - Headache, neurologic deficits, seizures
  - Multiple lesions seen at gray-white junction on cranial MRI
- Bone
  - Pathologic fractures
- Liver
  - Hepatomegaly
  - Jaundice
- Metastasis **TO** the lung (from other location): Breast, colon, kidney
  - Look for MULTIPLE LESIONS in the lung

Source: UpToDate, Amboss

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Lung Cancer Staging

Staging of NON-Small cell lung cancer (NSCLC)

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- Union for international cancer control (UICC) determined Lung Ca stage from the Tumor, Node, Metastasis (TNM) system:
  - (**T**) size and extent of the main **T**umor
  - (**N**) **N**umber of nearby lymph nodes involved
  - (**M**) presence of distal **M**etastasis

As soon as distant metastases (M1) are detected, the cancer is classified as UICC stage IV

Source: UpToDate, Amboss

Staging of SMALL Cell Lung Cancer (SCLC)

- Based primarily on whether tumor is limited to one hemithorax or has spread beyond the hemithorax

Classification	Cancer spread	Corresponding TNM
Very limited disease	Confined to one hemithorax	T1–2, N0–1
Limited disease		T3–4, N0–1 or T1–4, N2–3
Extensive disease	Beyond one hemithorax • Contralateral supraclavicular or hilar lymph node involvement • Malignant pericardial or pleural effusion • Distant metastasis	M1

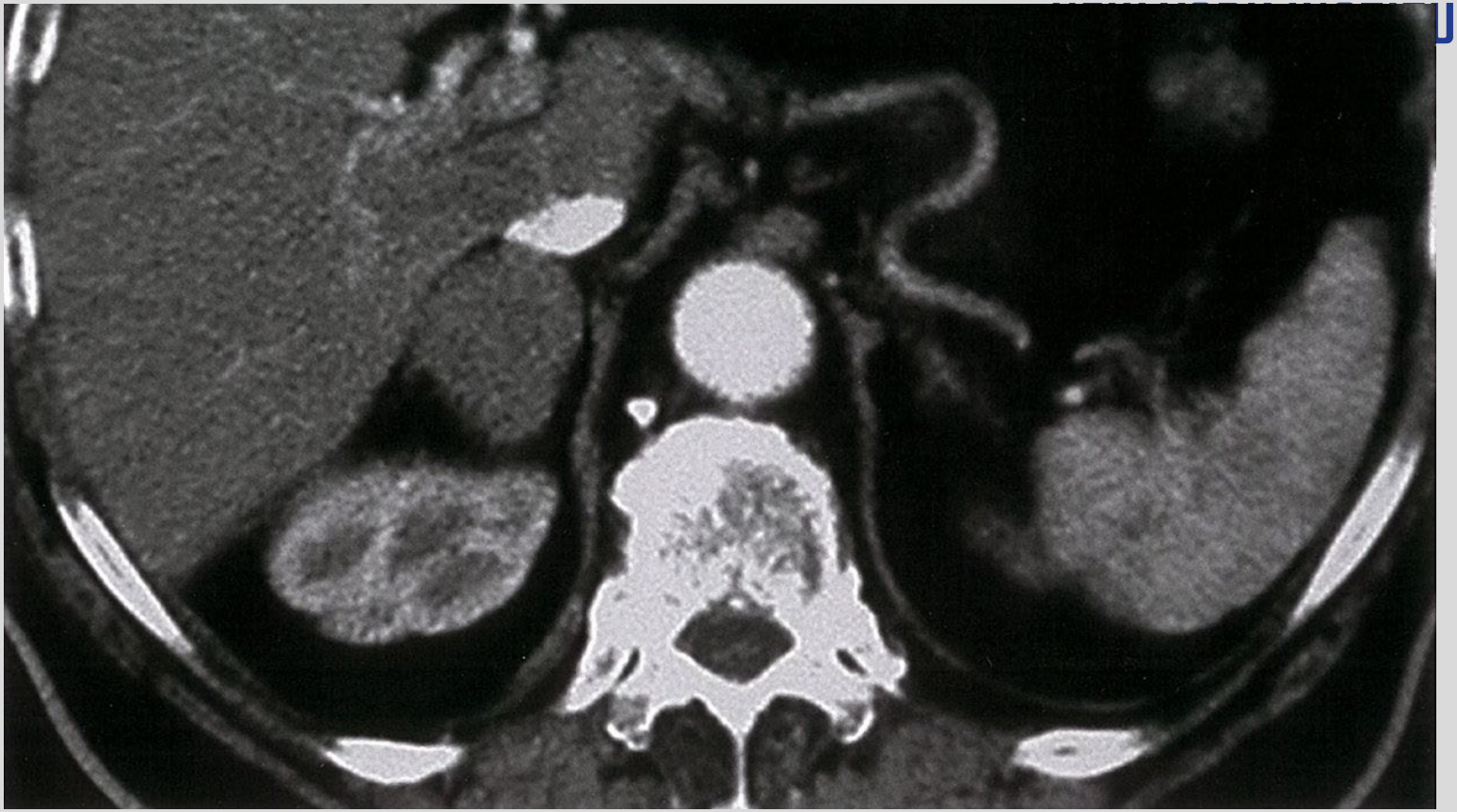
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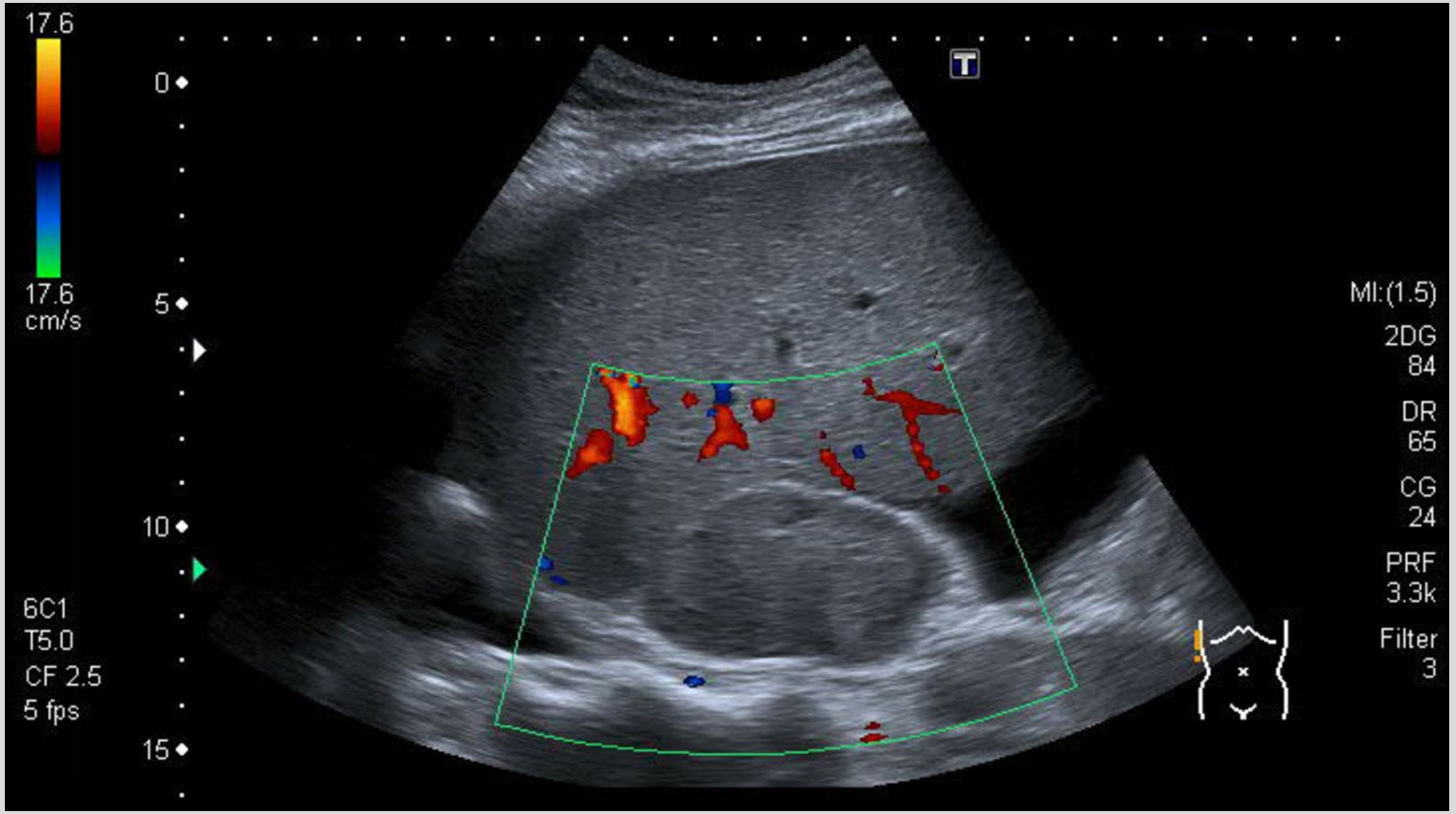
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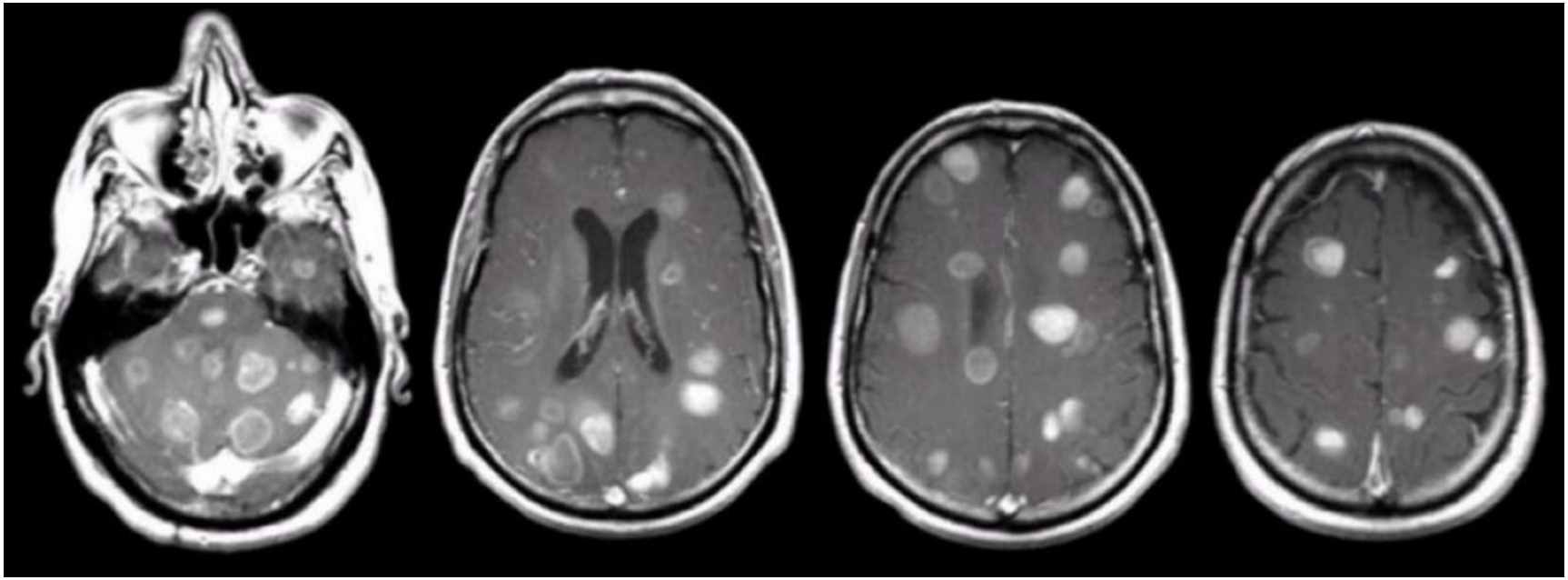
How do we stage lung cancer?

How to stage lung cancer

- CT scan (chest, liver, adrenal glands) to evaluate
  - Nodules: localization, size, margins, calcifications
  - Regional spread (hilar lymph nodes, mediastinal invasion, pleural effusions)
  - Distant metastasis (hepatic lesions, adrenal gland mass)
- Further radiographic imaging (based on clinical features and/or lab findings indicating metastatic disease)
  - Abdominal US and CT: assesses lymph node involvement and extent of metastatic disease
  - Skeletal scintigraphy: detects bone metastases
  - Cranial MRI: detects CNS metastases
  - Whole body PET-CT: best method for detection of occult disease




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What labs might we order?

- Complete blood count
- Electrolytes (including BUN/Cr)
- Calcium
- Alkaline phosphatase
- Alanine aminotransferase (ALT) and aspartate aminotransferase (AST), total bilirubin
- Albumin and lactate dehydrogenase (not essential)

Treatment of Lung Cancer

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NSCLC

- Stage 1, 2, 3a → Curative
  - Surgical resection +/- adjuvant chemo
  - If surgery not possible → chemo + radiation → consider sx if tumor shrinks
- Stage 3b, 4 → Palliative (symptom oriented)

SCLC

- Limited disease → curative
  - Chemo + radiation
  - Prophylactic cranial irradiation in patients who respond to initial chemo
- Extensive disease → palliative

Source: Amboss

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Early Palliative Care for Patients with Metastatic Non–Small-Cell Lung Cancer

Jennifer S. Temel, M.D., Joseph A. Greer, Ph.D., Alona Muzikansky, M.A., Emily R. Gallagher, R.N., Sonal Admane, M.B., B.S., M.P.H., Vicki A. Jackson, M.D., M.P.H., Constance M. Dahlin, A.P.N., Craig D. Blinderman, M.D., Juliet Jacobsen, M.D., William F. Pirl, M.D., M.P.H., J. Andrew Billings, M.D., and Thomas J. Lynch, M.D.

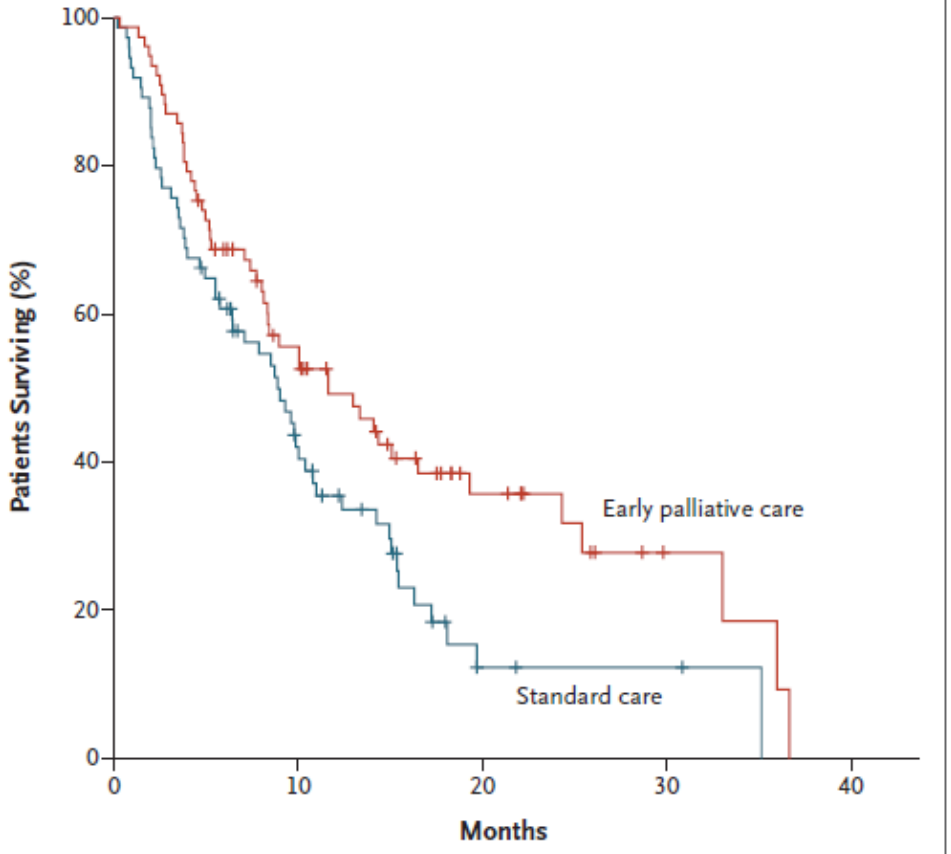


Figure 3. Kaplan–Meier Estimates of Survival According to Study Group.

Treatment of Lung Cancer

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Medical therapy

- Polychemotherapy e.g. SCLC: cisplatin and etoposide
- Targeted therapy when tumor has positive cytological markers: EGFR inhibitors, ALK tyrosine kinase inhibitors

Surgical management

- Open thoracic surgery VS. VATS
  - Lobectomy (standard approach): resection of one lobe
    - If FEV1 > 1.5 L and DLCO > 60%: lobectomy can be tolerated
  - Wedge resection for patients with small, localized tumors who cannot tolerate lobectomy
  - Pneumonectomy: complete lung resection in the case of a central tumor

Source: UpToDate, Amboss

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How do we screen for lung cancer?

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ESTABLISHED IN 1812 AUGUST 4, 2011 VOL. 365 NO. 5

Reduced Lung-Cancer Mortality with Low-Dose Computed Tomographic Screening

The National Lung Screening Trial Research Team*

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METHODS

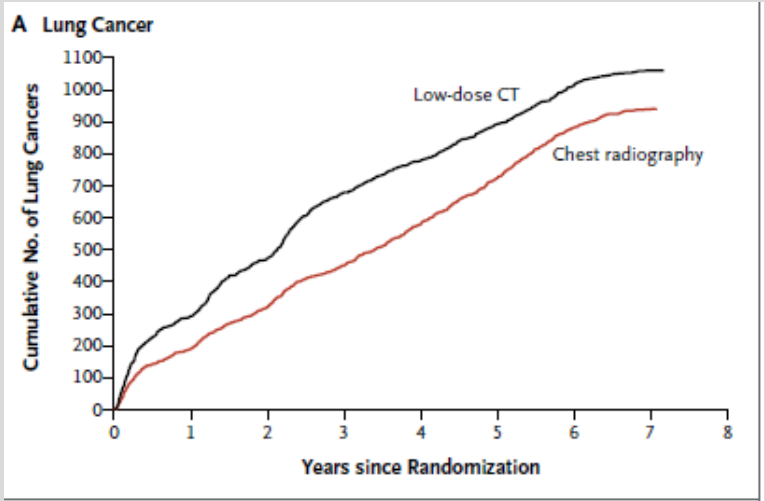
From August 2002 through April 2004, we enrolled 53,454 persons at high risk for lung cancer at 33 U.S. medical centers. Participants were randomly assigned to undergo three annual screenings with either low-dose CT (26,722 participants) or single-view posteroanterior chest radiography (26,732). Data were collected on cases of lung cancer and deaths from lung cancer that occurred through December 31, 2009.

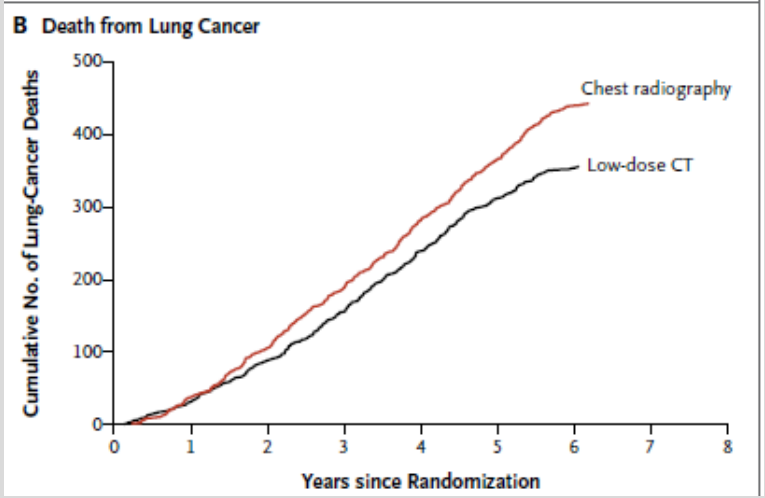
RESULTS

The rate of adherence to screening was more than 90%. The rate of positive screening tests was 24.2% with low-dose CT and 6.9% with radiography over all three rounds. A total of 96.4% of the positive screening results in the low-dose CT group and 94.5% in the radiography group were false positive results. The incidence of lung cancer was 645 cases per 100,000 person-years (1060 cancers) in the low-dose CT group, as compared with 572 cases per 100,000 person-years (941 cancers) in the radiography group (rate ratio, 1.13; 95% confidence interval [CI], 1.03 to 1.23). There were 247 deaths from lung cancer per 100,000 person-years in the low-dose CT group and 309 deaths per 100,000 person-years in the radiography group, representing a relative reduction in mortality from lung cancer with low-dose CT screening of 20.0% (95% CI, 6.8 to 26.7; P=0.004). The rate of death from any cause was reduced in the low-dose CT group, as compared with the radiography group, by 6.7% (95% CI, 1.2 to 13.6; P=0.02).

CONCLUSIONS

Screening with the use of low-dose CT reduces mortality from lung cancer. (Funded by the National Cancer Institute; National Lung Screening Trial ClinicalTrials.gov





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Prevention & Screening

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- Cessation of smoking
  - After cessation, the risk of lung cancer reduces by ½ within 5–10 years.
  - After 15–20 yrs risk continues to decrease but always remains slightly elevated vs a non-smoker.
- **Screening 2025**
  - low-dose CT in patients aged 50–80 (previously 55-74) years (USPSTF recommends 55–80 years) with either:
  - ≥ 20 pack yrs (previously 30) in current smokers or those who've stopped within 15 years

Source: UpToDate, Amboss

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Thank you

Lecture Feedback Form:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R